



Effect of external and internal plasticization on the glass transition temperature of (Meth)acrylate polymers studied with molecular dynamics simulations and calorimetry



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HIGHLIGHTS

- Integrating a free plasticizer into a (meth)acrylate terpolymer increases the glass transition temperature.
- Plasticizers lower polymer glass transition by inducing frequently small displacements of polymer segments.
- Bulky internal plasticizers effectively lower glass transition by weakening electrostatic interactions within polymer.
- Mobility of compact free plasticizers lower glass transition by inducing polymer segment displacements more frequently.

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ABSTRACT

The differences between internal and external plasticization as well as how the molecular structure of a plasticizer effects the glass transition temperature, T_g , of a (meth)acrylate terpolymer is analyzed with molecular dynamics (MD) simulations and measured with differential scanning calorimetry (DSC). 1-phenoxy-2-propanol (PPH), butyl carbitol (BC) and ethylene glycol-2-ethylhexyl ether (EEH) are added either as free external plasticizers or are integrated into the polymer as internal bonded plasticizers. The simulation model is validated by finding excellent agreement of predicted T_g -values of various relevant homopolymers and terpolymers with measured values and values that are derived from established empirical equations. Simulation results suggest an aggregation of polymer alkyl groups into non-polar nano-domains. Moreover, addition of the three external plasticizers lowers T_g substantially. This T_g -loss is partially recovered after directly integrating the free plasticizers into the polymer. Bulkier plasticizers, such as BC, are more effective internal plasticizers because their larger size increase the distance between polymer chains to a larger extend, which leads to weaker average interactions among the chains, thereby reducing T_g . On the other hand, more compact free plasticizers, such as PPH, exhibit a lower diffusive energy barrier that causes more frequent diffusive moves. This stimulates the diffusive movement also of the polymer, which reduces T_g . Hence, integrating such a compact plasticizer into the polymer leads to a larger recovery of the T_g - loss that was induced by its free form. Overall, the complementary use of MD simulations and DSC measurements provide insights on the differences on how internal and external plasticizers influence T_g , respectively, and identifies the underlying molecular causes. External plasticization reduces T_g more effectively than internal plasticization. Hence, a drop in T_g caused by an external plasticizer can be partially recovered by integrating the plasticizer into the polymer. A possible application is the design of reactive coalescing agents for coatings that bind to the polymer after application and that could replace conventional and potentially harmful coalescing agents that evaporate into the environment.

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